

RCS (Radar Cross Section) Tool Software User Manual

1. Introduction

1.1 Purpose

The Radar Cross Section tool is designed to allow users to create complex geometries on top of SAR (synthetic aperture radar) data and calculate the RCS (radar cross section) as well as other related parameters for the selected geometries. This tool was created to update and improve legacy tools, using SARPpy for most of the backend SAR related tasks.

1.2 Document Overview

This Software User Manual (SUM) is structured to provide users with all the necessary information to install and effectively use the RCS tool. The manual is organized as follows:

1. **Introduction:** An overview of the software's purpose and design
2. **Installation Instructions:** Step-by-step guidance for setting up the software
3. **Graphical User Interface:** Detailed explanation of the interface and navigation
4. **Model View Controller:** Technical overview of the software design pattern used
5. **Conclusion:** Documentation summary
6. **References:** Citations for references.

2. Installation Instructions

Python Install

1. Navigate to the main rcs-tool directory
2. pip install -r requirements.txt
3. pip install src/.
4. Install PyMITM

Standalone Application

PyRCS can be packaged as a standalone executable using PyInstaller:

1. Install PyMITM if not already installed:

2. Navigate to the PyInstaller directory

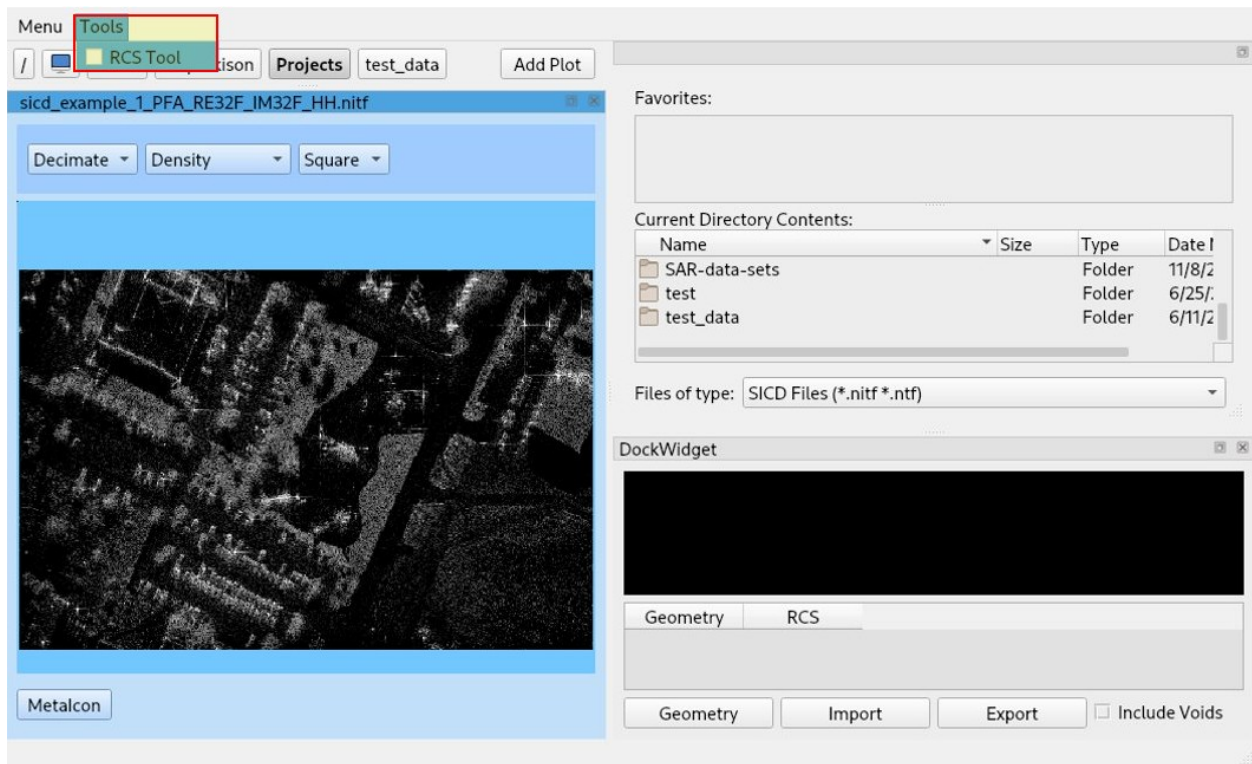
3. Create the executable:

```
pyinstaller mitm.spec
```

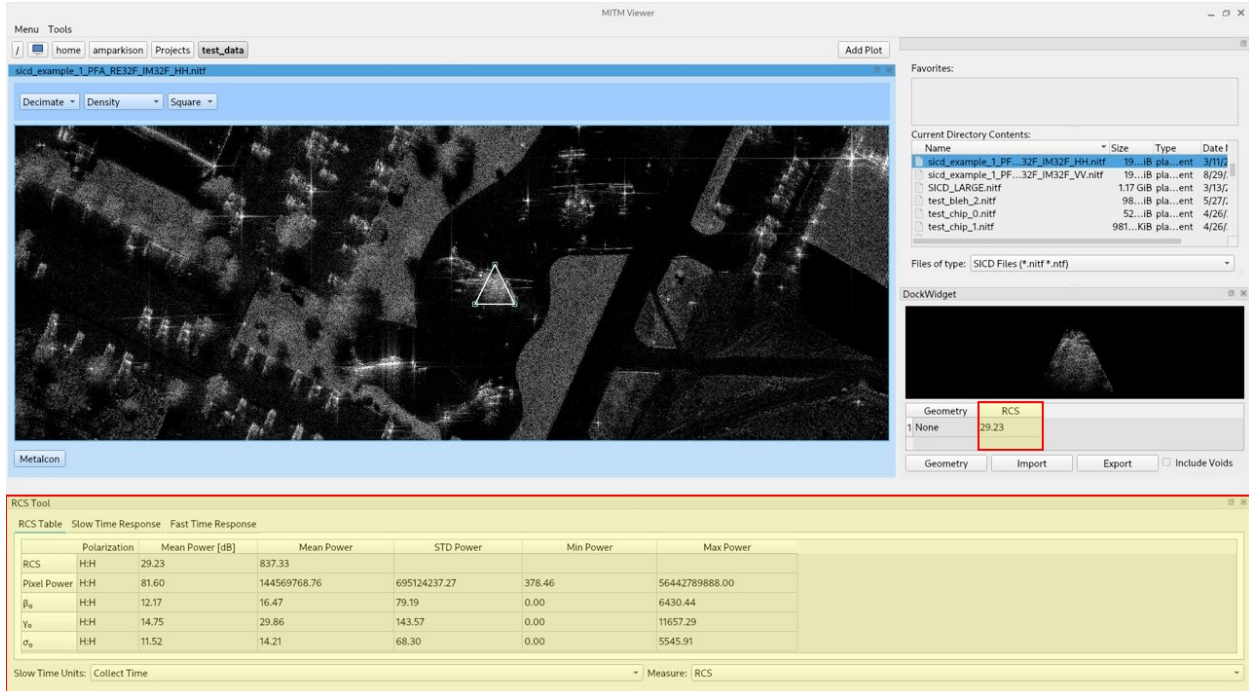
4. Run the executable:

```
./dist/mitm.exe
```

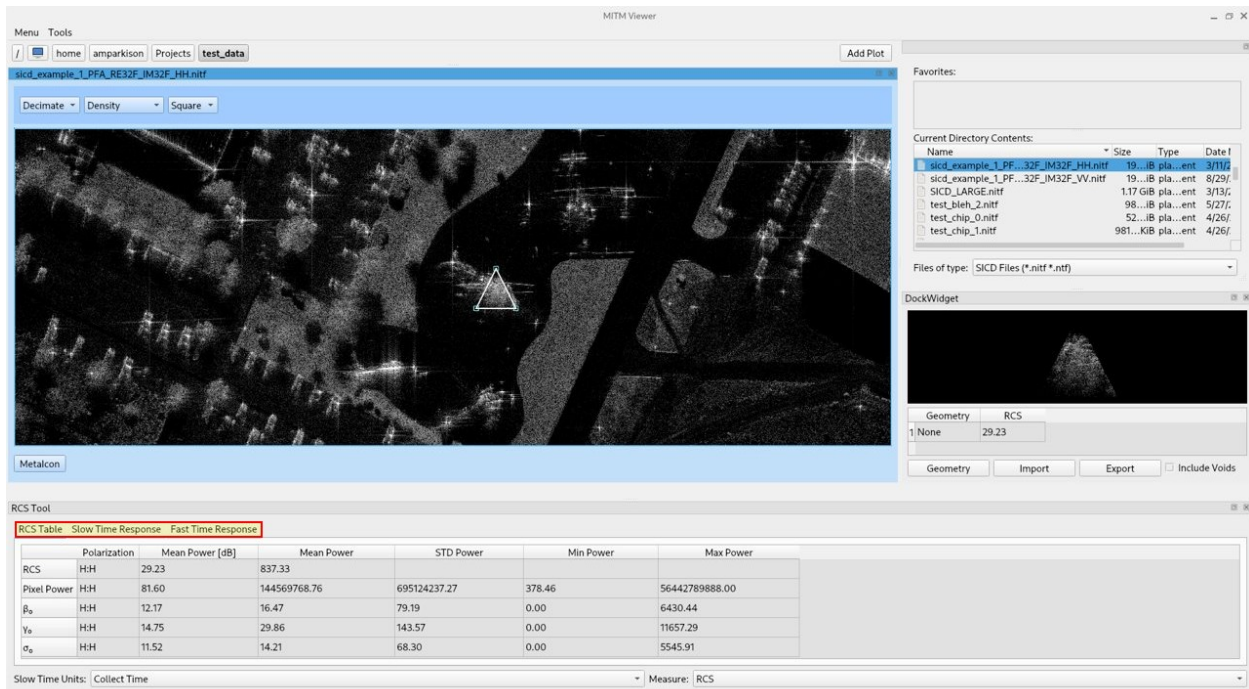
3. Graphical User Interface (GUI)



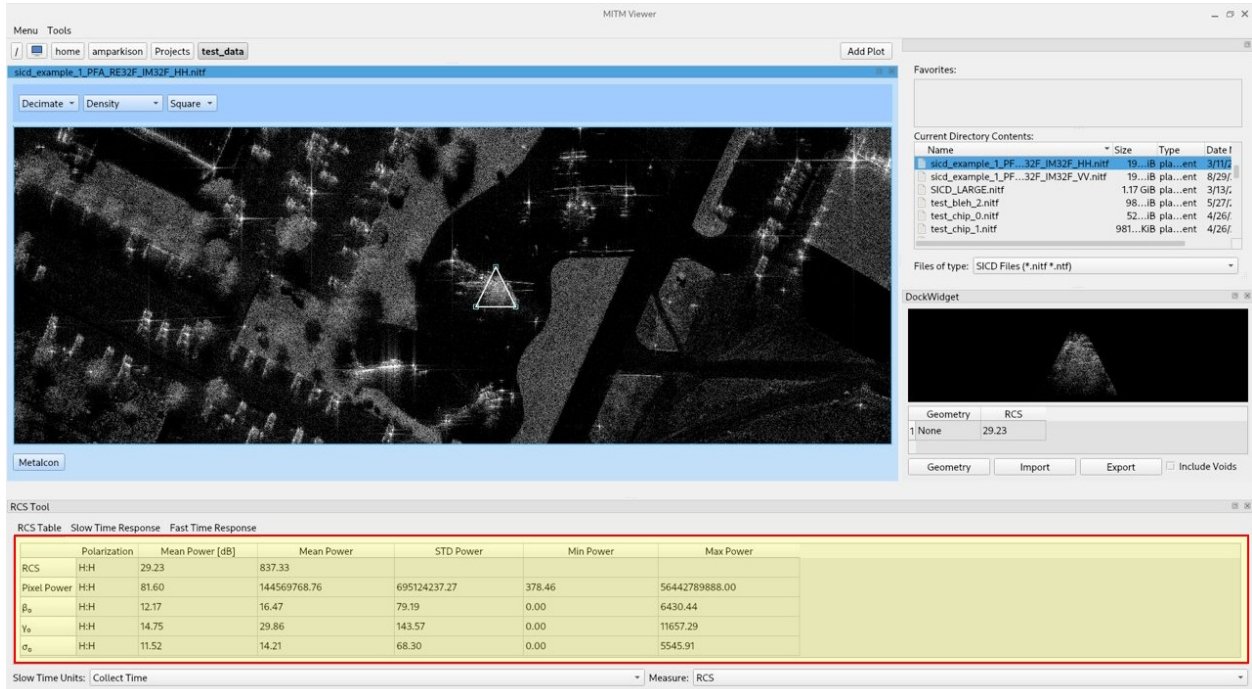
The RCS Tool exists as one main dock widget that gets added to MITM Viewer, by default this is displayed to the bottom of MITM Viewer but can be moved around and docked and undocked. To activate the RCS Tool dock widget the user can go to the tools menu in MITM Viewer and check the "RCS Tool".



The RCS Tool is displayed below MITM Viewer as it's own dock widget. The RCS Tool also extends some features of MITM Viewer as you can see a RCS header is added to the geometry table of MITM Viewer. This will display the current selected geometries RCS parameter depending on currently selected measure parameter.



The main windows for displaying information to the user are controlled by three tabs: **RCS Table**, **Slow Time Response**, and **Fast Time Response**. The user can click these different tabs to change the view of the displayed data in the region below the tabs.



RCS Table: Displays a table of various calculated RCS parameters relating to the currently selected geometry. These parameters consist of **RCS**, **Pixel Power**, **Beta Naught**, **Gamma Naught**, and **Sigma Naught**.

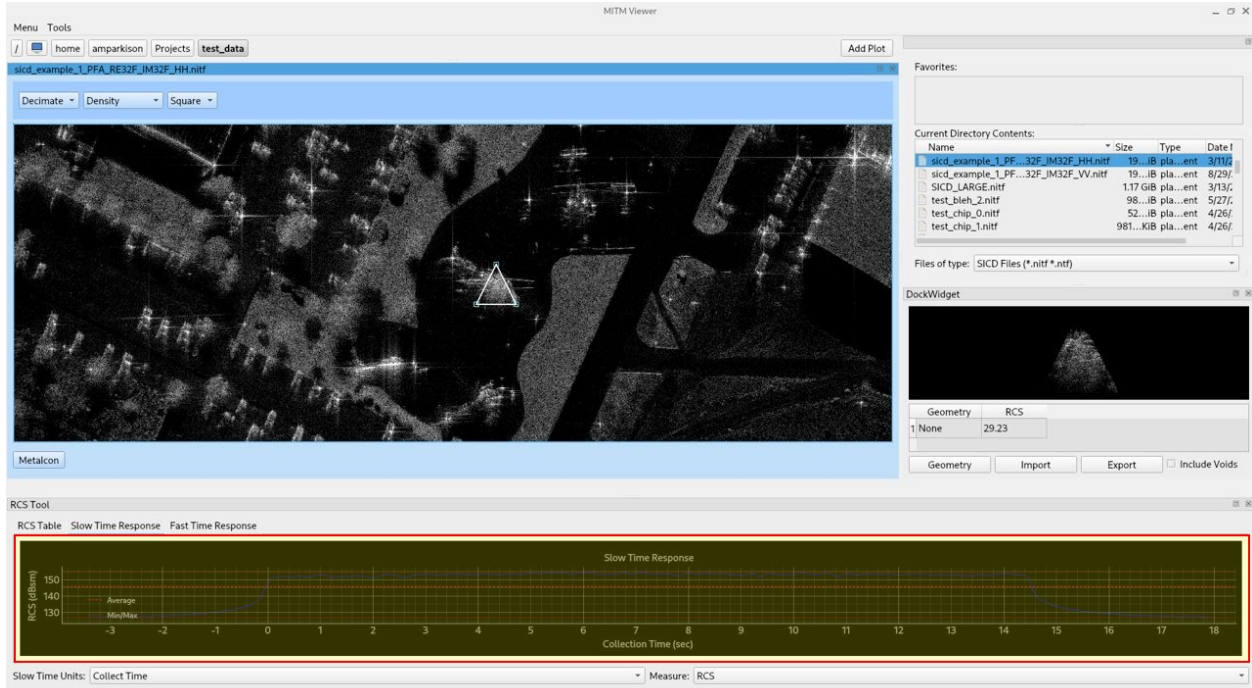
RCS (Radar Cross Section): A measure of how detectable an object is by radar - essentially how much radar energy bounces back toward the receiver.

Pixel Power: The total back-scattered energy received by a SAR sensor for each pixel in the image.

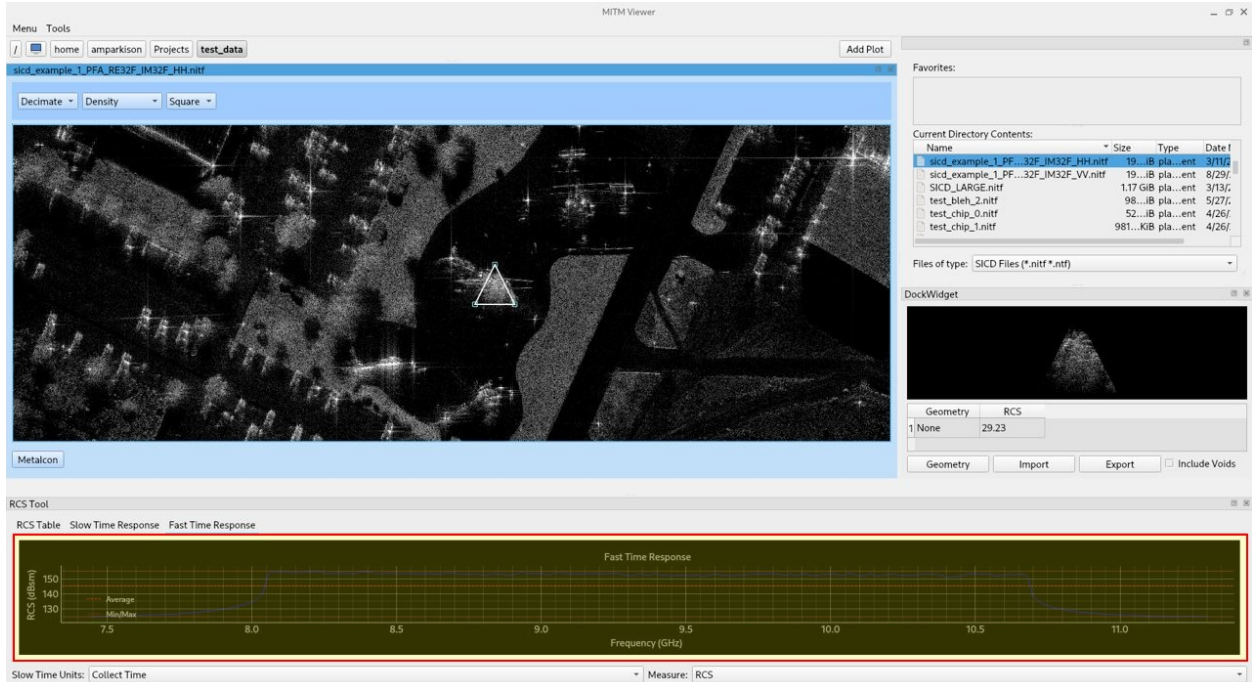
Beta Naught (β^0): Radar brightness coefficient normalized per unit area in slant range geometry. Represents back-scattering per unit area as the radar "sees" it, accounting for the slanted viewing angle.

Gamma Naught (γ^0): Radar brightness coefficient normalized per unit area on the tilted ground surface. Accounts for local topographic slope effects on back-scattering.

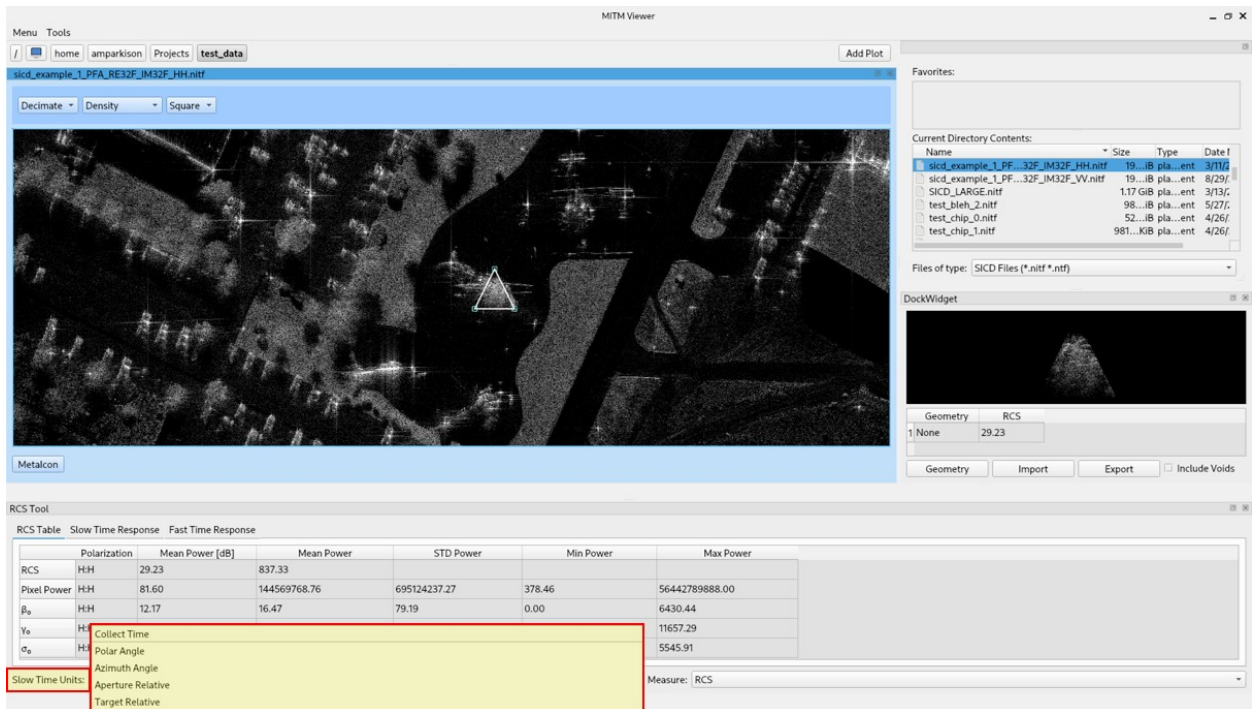
Sigma Naught (σ^0): Radar back-scattering coefficient normalized per unit area of flat ground surface. Representing the intrinsic scattering properties of the target independent of viewing geometry.



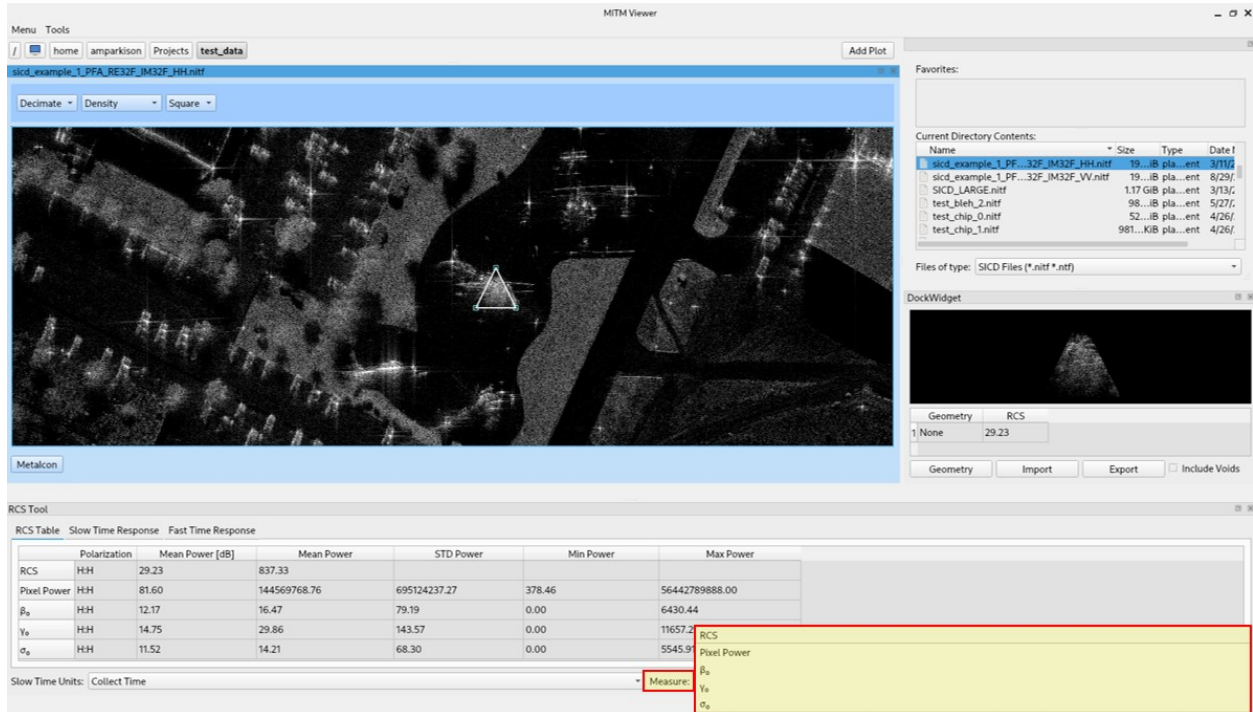
Slow Time Response: Displays a plot of the slow time response. The user can change what is plotted by selecting different slow time units and measure parameters in the combo boxes below.



Fast Time Response: Displays a plot of the fast time response. The user can change the y axis measure parameter by selecting an option from the measure below the plot.



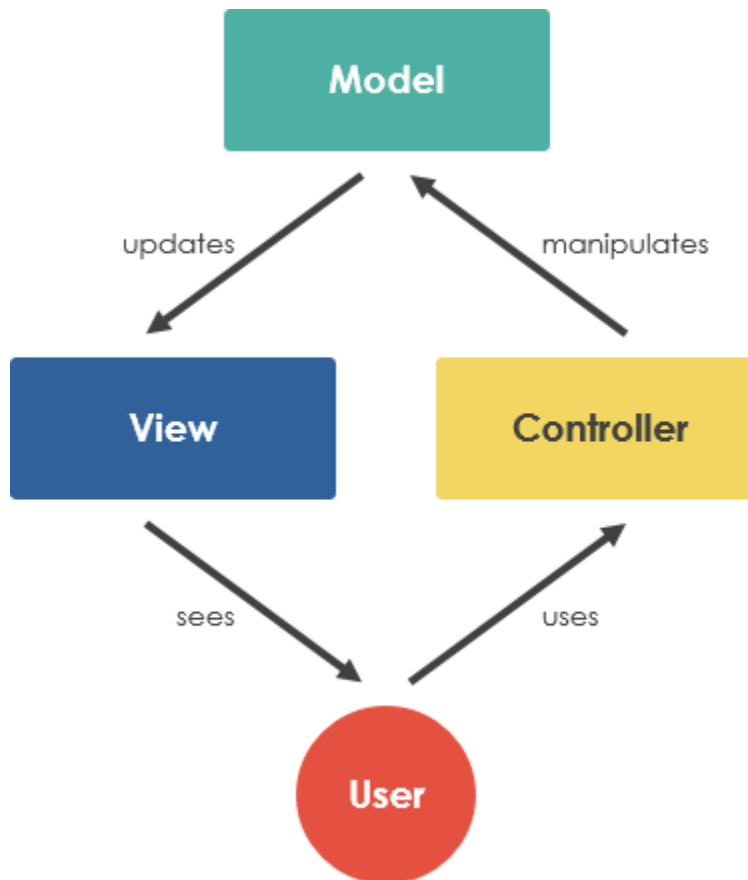
Slow Time Units: This combo box allows the user to select the slow-time units. This impacts the slow time response plot.

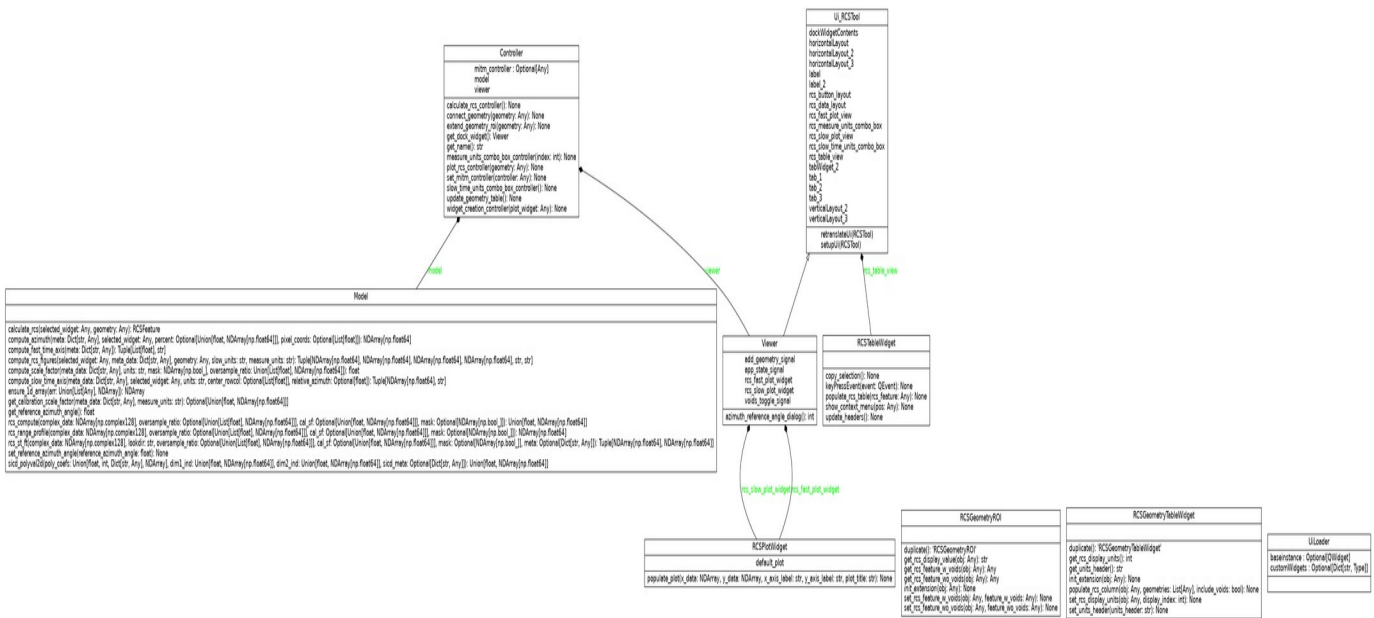


Measure: This combo box allows the user to select the measured parameter. This impacts the slow time response plot, fast time response plot, and the geometry table (the currently displayed value relating to the RCS will be whatever parameter the user has selected here).

4. Model View Controller (MVC)

The RCS-tool was made following the MVC (Model-View-Controller) design pattern. MVC separates the application in three main components. The model and viewer are completely isolated from one another, the model contains all of the brains of the application and functionality, the viewer contains all of the GUI elements and handles user interaction. The controller marries the model and viewer so that as a user interacts with the GUI any back-end calculations or logic can be completed by the model by having the controller send the input data from the viewer to the model and then back to the viewer to update the information to the user.





In conclusion the RCS tool gives users the ability to create complex geometries on complex SAR data shown as images using MITM viewer and calculate the RCS and related parameters for in depth analysis and aids in organizing this data in a presentable manner to the user. The RCS tool effectively enhances and replaces former legacy tools.

6. References

[1] Visual Paradigm. From Use Case to MVC Framework: A Guide Object-Oriented System Development. [Online]. Available: <https://guides.visual-paradigm.com/from-use-case-to-mvc-framework-a-guide-object-oriented-system-development/>. [Accessed: March 31, 2025].